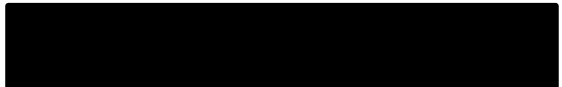
Base conversion (Decimal, binary, octal, hexadecimal)



All relevant bases of number systems:

Binary: 0, 1

Octal: 0, 1, 2, 3, 4, 5, 6, 7

Decimal: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

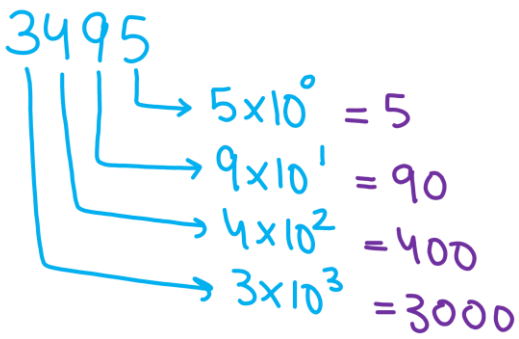
Hexadecimal: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F

What is base of a number system?

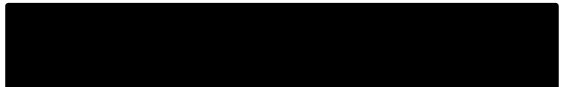
- The number of digits used to create counting

Place value is decided

Example:



How are numbers generated?

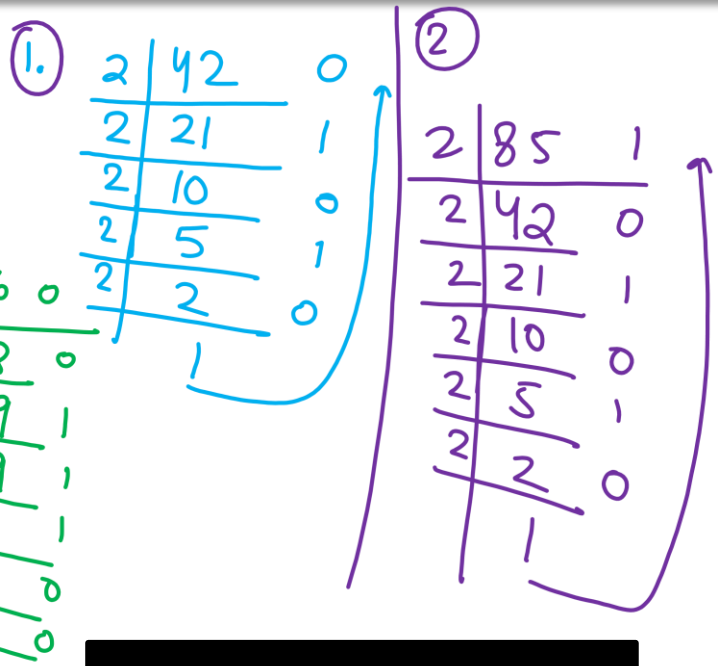


Decimal to binary and vice versa

1) $42_{10} = (101010)_2$

2) $85_{10} = (1010101)_2$

3) $156_{10} = (10011100)_2$



Binary to decimal, last question!

1. $(101010)_2 = (42)_{10}$

$2^5 + 2^3 + 2^1$

$32 + 8 + 2 = 42$

2. $(1010101)_2 = (85)_{10}$

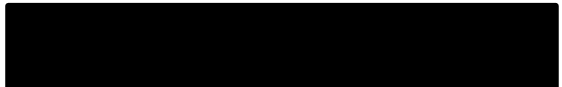
$2^6 + 2^4 + 2^2 + 2^0 = 64 + 16 + 4 + 1 = 85$

3. $(10011100)_2 = (156)_{10}$

$2^7 + 2^4 + 2^3 + 2^2 = 156$

1. **Converting the decimal number 241 to the binary number system we get:**

a) 11110000
b) 11110001
c) 10110001
d) 11010001



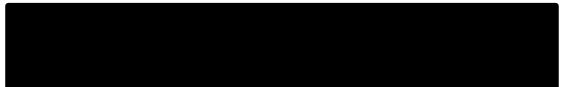
2. **The binary number 11001 in the decimal number system is equal to:**

a) 23
b) 24
c) 25
d) 26

[redacted]

3. **Binary number 11111111 added to 10001000 becomes decimal number:**

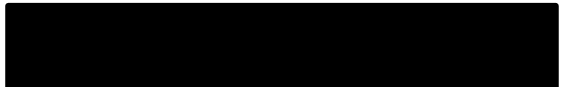
a) 373
b) 374
c) 391
d) 376



4. **The binary number 11111111 divided by 1111 gives _____ in the decimal system.**

a) an even number as the answer
b) a composite number as the answer
c) the highest prime number less than 20
d) the second highest prime number less than 20

a)
b)
c) 20
d) 20

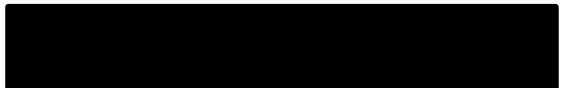


Decimal to Octal and vice versa

1) 59_{10}

2) 120_{10}

3) 345_{10}



Octal to decimal, last page ques:

① 73_8

8^1 8^0

$= 56 + 3 = 59_{10}$

② 170_8

8^2 8^1 8^0

$64 + 56 = 120_{10}$

③ $531_8 =$

8^2 8^1 8^0

$320 + 24 + 1 = 345_{10}$

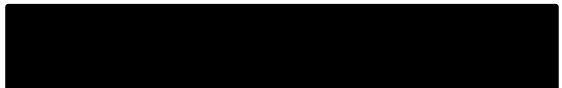
[redacted]

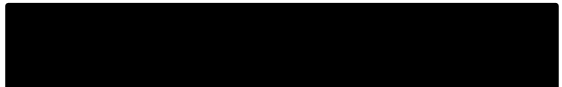
Decimal to Hexadecimal and vice versa

1) 78_{10}

2) 200_{10}

3) 1023_{10}





Convert the following numbers to binary, octal and hexadecimal numbers:

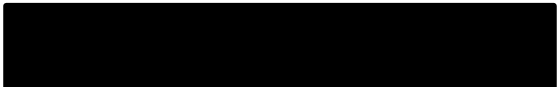
1) 25_{10}

2) 58_{10}

3) 127_{10}

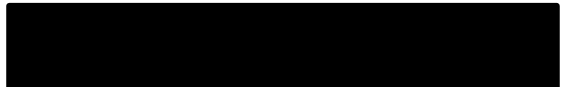
4) 255_{10}

5) 345_{10}

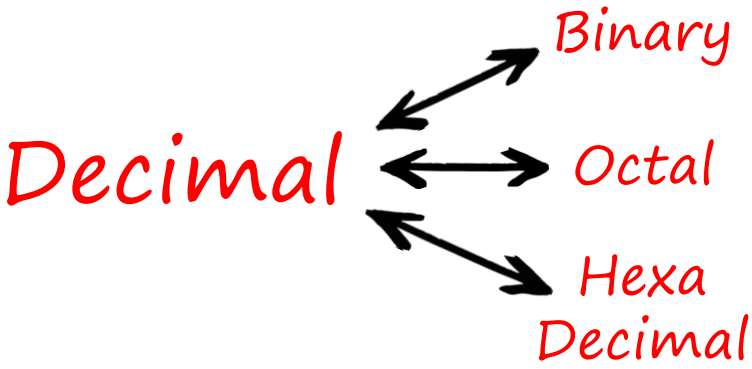


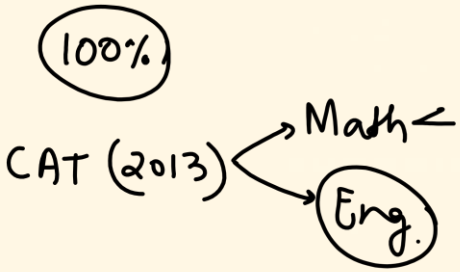
Answer key for last question:

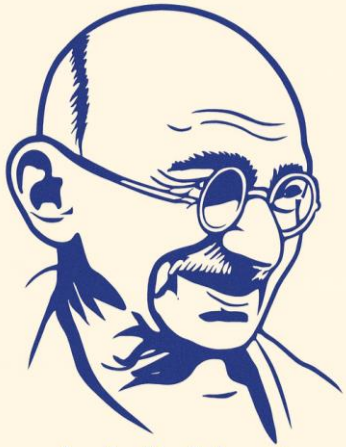
Decimal (base-10)	Binary (base-2)	Octal (base-8)	Hex (base-16)
25	11001	31	19
58	111010	72	3A
127	1111111	177	7F
255	11111111	377	FF
345	101011001	531	159



What if I have to convert from binary to octal etc:







"If I have the belief that I can do it, I shall surely acquire the capacity to do it even if I may not have it at the beginning."